Project Gaja-Saviya

An Intelligent Multi-Modal Framework for Human–Elephant Conflict Mitigation

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Individual Project Proposal Report

Component: Cross-Modal Sensor Fusion

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Human–Elephant Conflict (HEC) in Sri Lanka imposes severe socioeconomic burdens on rural farming communities, with existing detection systems suffering from high false alarm rates due to their inability to integrate and validate signals from multiple sensor modalities. This research proposes a Cross-Modal Sensor Fusion component as part of Project Gaja-Saviya, an intelligent AI-driven IoT framework for HEC mitigation. The proposed component acts as the central decision-making engine that ingests outputs from the electrical signal interpreter (LSTM-based fence anomaly detection) and the bio-acoustic behavior analyst (CNN-based elephant sound classification), synchronizes them temporally, and applies probabilistic fusion techniques to compute a validated threat confidence score. By combining rule-based filtering with Bayesian inference methods, the system ensures alerts are raised only when corroborating evidence from independent sensor streams concurs, significantly reducing false positives and improving situational reliability. The validated threat output, enriched with severity metadata, is then forwarded to the mobile application layer for real-time farmer guidance. This research contributes a robust, scalable, and context-aware sensor fusion mechanism that enhances the trustworthiness of HEC alert systems.

Keywords: Cross-Modal Fusion, Human–Elephant Conflict, Sensor Fusion, Bayesian Inference, IoT, Decision Support

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
IoT	Internet of Things
HEC	Human–Elephant Conflict
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
ML	Machine Learning
API	Application Programming Interface
ESP32	Espressif Systems 32-bit Microcontroller
RNN	Recurrent Neural Network
MQTT	Message Queuing Telemetry Transport
GPS	Global Positioning System
UI	User Interface
UX	User Experience
REST	Representational State Transfer

1. INTRODUCTION

1.1 Background and Context

Human–Elephant Conflict (HEC) is a longstanding and increasingly severe problem in Sri Lanka's rural agricultural zones. Elephants routinely breach farm perimeters, causing extensive crop damage, property destruction, and in many cases, threats to human life. According to the Department of Wildlife Conservation, Sri Lanka records dozens of human fatalities and hundreds of elephant deaths annually as a consequence of these conflicts. Despite governmental and civil efforts, the problem persists due to a lack of intelligent, adaptive, and reliable early warning systems.

Existing mitigation mechanisms, such as electric fences and acoustic deterrents, operate in isolation and are incapable of contextual reasoning. They trigger alarms based on single-source inputs — a voltage drop in a fence or a detected sound — without verifying the cause through corroborating evidence. This inability to cross-validate sensor data results in high false alarm rates, which erode farmer trust and lead to alert fatigue. What is needed is an intelligent middleware layer that aggregates, synchronizes, and validates information from multiple sensor sources before issuing any alert.

Project Gaja-Saviya addresses this gap by proposing a layered AI-IoT framework comprising four specialized components. This proposal focuses on the Cross-Modal Sensor Fusion component, which serves as the intelligent decision hub of the system. It receives processed inferences from two upstream modules — the electrical signal interpreter and the bio-acoustic behavior analyst — and fuses their outputs using probabilistic methods to issue validated, context-aware threat decisions. This component is architecturally central: without it, the system's multi-modal sensing capability cannot translate into reliable alerts.

1.2 Literature Review

Sensor fusion is a well-established field in robotics, autonomous vehicles, and surveillance systems, where data from heterogeneous sources are integrated to produce more reliable inferences than any single source could provide. Early foundational work by Hall and Llinas (1997) introduced the concept of multi-sensor data fusion as a systematic process for combining sensor data and related information from different sources to achieve improved accuracy and inferences.

In wildlife monitoring contexts, researchers have explored acoustic and camera-based detection independently, but few have implemented multi-modal fusion specifically for HEC. Studies using passive acoustic monitoring (PAM) have demonstrated the effectiveness of identifying elephant vocalizations from environmental audio. Concurrently, IoT-based electric fence monitoring systems have been deployed in South Asia and Africa, relying on voltage anomaly detection. However, these systems lack integration.

Bayesian inference has emerged as a preferred method for decision fusion because it provides a principled way to combine uncertain evidence from multiple sources using prior knowledge and conditional probabilities. Dempster-Shafer theory offers an alternative that handles uncertainty without requiring precise prior distributions, and has been applied in intrusion detection systems with promising results.

Recent work in AI-driven environmental monitoring has combined LSTM networks for temporal signal analysis with CNN-based spectrogram classifiers, and applied fusion at the decision layer to produce more robust outputs. Closest to the proposed architecture is the work by Sharma et al. (2023), who demonstrated that combining vibration and acoustic sensors for perimeter security using a Bayesian fusion layer reduced false alarms by over 40% compared to single-modal approaches. Project Gaja-Saviya builds upon these insights, adapting them for the specific challenges of elephant intrusion in tropical agricultural contexts.

1.3 Research Gap

The literature reveals several critical gaps relevant to this component. First, while individual sensor modalities for elephant detection exist, no integrated cross-modal fusion system has been specifically designed and evaluated for HEC in Sri Lanka. Second, existing fusion methods in wildlife monitoring contexts lack real-time processing constraints required for farmer alert systems, where latency must be minimized. Third, the challenge of temporal misalignment between heterogeneous sensor streams — electrical signals arriving at millisecond granularity versus acoustic frames processed at second intervals — has not been addressed in HEC-specific solutions. Fourth, current systems do not implement trust-weighted fusion, which would allow the system to adaptively rely more on sensors that have demonstrated higher reliability under specific environmental conditions. This research addresses all four gaps within the Gaja-Saviya framework.

2. OBJECTIVES

2.1 Main Objective

To design and implement a cross-modal sensor fusion engine that integrates electrical fence anomaly signals and bio-acoustic elephant behavior inferences in real time, producing validated, confidence-scored threat alerts with minimized false positives for use in HEC mitigation.

2.2 Specific Objectives

1. To develop a temporal synchronization mechanism that aligns outputs from the electrical signal interpreter and the bio-acoustic analyzer within a defined time window for coherent fusion.
2. To implement a probabilistic decision fusion algorithm (Bayesian and/or rule-based) that computes a joint threat confidence score from multi-modal sensor inputs.
3. To design a severity classification schema that categorizes validated threats into levels (low, medium, high) based on combined sensor evidence.
4. To build a filtering mechanism that suppresses duplicate or temporally clustered alerts to avoid alarm flooding during extended intrusion events.
5. To evaluate the fusion engine's performance against single-modal baselines using metrics of precision, recall, false positive rate, and alert latency.
6. To ensure seamless API integration between the fusion engine and the mobile application layer for real-time alert delivery.

3. METHODOLOGY

This research adopts a Design Science Research (DSR) methodology, emphasizing the iterative design, construction, and evaluation of a functional artifact — the cross-modal sensor fusion engine. The methodology proceeds through four primary phases.

Phase 1: Data Collection and Interface Definition

The fusion engine requires structured inputs from two upstream components. During this phase, the input schemas from the electrical signal interpreter (disturbance type, confidence score, timestamp) and the bio-acoustic analyzer (behavior label, confidence score, timestamp) will be formally defined and standardized. Controlled lab experiments will generate paired datasets representing synchronized multi-modal events across multiple intrusion scenarios: true elephant contact, environmental noise, branch impact, and equipment malfunction.

Phase 2: Temporal Synchronization Module

A sliding time window mechanism will be implemented to pair and align asynchronous sensor outputs. Given that electrical signals are sampled at high frequency (500 Hz) while acoustic analysis operates on 1–5 second frames, the synchronizer will buffer recent predictions from both streams and trigger fusion evaluation whenever both modalities produce an output within a defined co-occurrence window (configurable, default 3 seconds). Timestamps will be normalized using NTP-synchronized clocks on the ESP32 nodes.

Phase 3: Fusion Algorithm Design

Two complementary fusion strategies will be implemented and compared. The first is a rule-based fusion layer that applies expert-defined logical rules (e.g., 'Alert if electrical confidence > 0.7 AND acoustic detects aggressive behavior'). This ensures baseline reliability and interpretability. The second is a Bayesian probabilistic fusion model that computes the posterior probability of a genuine elephant intrusion given the joint sensor evidence, using prior probabilities derived from the training dataset. The final system will combine both, using rule-based gating to trigger initial evaluation and Bayesian scoring to compute nuanced confidence.

Phase 4: Output, Filtering, and Integration

The fusion engine will produce a validated threat object comprising: threat status (boolean), severity level (low/medium/high), joint confidence score (0–1), and metadata (timestamps, contributing sensor IDs). A short-term alert buffer will suppress repeated alerts from the same event within a 5-minute window. A RESTful API endpoint will expose the fusion output to the mobile application layer. The engine will be implemented in Python, deployed on the backend server, and tested with both simulated and real sensor data.

4. SYSTEM ARCHITECTURE

4.1 Component Architecture Overview

The Cross-Modal Sensor Fusion component occupies the middleware position in the Gaja-Saviya architecture, connecting the IoT sensing layer to the mobile application layer. It receives structured prediction objects from the Electrical Signal Interpreter (upstream) and the Bio-Acoustic Behavior Analyst (upstream) via internal API calls, and forwards validated threat objects to the Dynamic Safe-Routing and Trust Platform (downstream) via a REST webhook.

4.2 Internal Sub-Components

- Input Receiver & Validator: Validates incoming prediction JSON objects against defined schemas and rejects malformed inputs.
- Temporal Synchronizer: Maintains a rolling buffer of recent predictions and pairs inputs within the co-occurrence time window.
- Rule Engine: Applies configurable logical rules to paired inputs to produce an initial alert flag.
- Bayesian Scorer: Computes the posterior threat probability using learned conditional distributions.
- Alert Buffer: Prevents alert flooding by deduplicating events within a sliding 5-minute window.
- Output Publisher: Serializes the validated threat object and dispatches it via REST API to the mobile layer and stores it in the database.

4.3 Data Flow

Electrical Interpreter output → Input Receiver → Temporal Synchronizer → [Rule Engine + Bayesian Scorer] → Alert Buffer → Output Publisher → Mobile Application Layer + Database

5. USER REQUIREMENTS

5.1 Stakeholders

- Primary Users: Farmers in HEC-affected regions who receive alerts from the mobile application downstream of this component.
- Technical Operators: System administrators responsible for configuring fusion thresholds and monitoring engine performance.
- Upstream Components: The Electrical Signal Interpreter and Bio-Acoustic Analyzer teams whose outputs feed into this component.
- Downstream Component: The Dynamic Safe-Routing and Trust Platform that consumes validated fusion outputs.

5.2 Functional Requirements

7. The system shall accept structured JSON prediction objects from both upstream sensor components via internal API calls.
8. The system shall synchronize temporally misaligned sensor outputs using a configurable sliding time window.
9. The system shall compute a joint threat confidence score using Bayesian fusion on paired sensor predictions.
10. The system shall apply rule-based filtering to suppress obviously false events before probabilistic scoring.
11. The system shall classify validated threats into severity levels: Low, Medium, and High.
12. The system shall prevent duplicate alerts for the same intrusion event within a configurable suppression window.
13. The system shall publish validated threat objects to the downstream mobile platform via REST API within 2 seconds of sensor input receipt.
14. The system shall log all fusion events (inputs, decisions, outputs) to a persistent database for audit and retraining.

5.3 Non-Functional Requirements

Requirement	Specification
Latency	End-to-end fusion decision within 2 seconds of paired input arrival
Accuracy	False positive rate below 10% on validation dataset
Availability	99.5% uptime for backend fusion service
Scalability	Support up to 50 concurrent sentinel nodes
Security	Encrypted API communication (HTTPS/TLS)
Maintainability	Modular design enabling threshold updates without redeployment

6. BUDGET AND JUSTIFICATION

Category	Amount (LKR)
Cloud Server (Backend Hosting)	18,000
Software Tools & Libraries	8,000
Dataset Acquisition & Storage	6,500
Testing & Validation	5,000
Contingency	4,000
TOTAL	**41,500**

Cloud server hosting constitutes the primary cost, as the fusion engine requires a reliable, always-on backend for real-time processing. Software tools include Python libraries (scikit-learn, scipy, Flask/FastAPI) which are open-source but require a licensed development environment. Dataset costs cover simulated sensor data generation and storage. Testing costs include equipment time for field validation with the Sentinel Node hardware.

7. WORK BREAKDOWN STRUCTURE

Phase	Task	Duration
1	Proposal Preparation & Literature Review	Months 1–2
2	Interface & Schema Definition with Upstream Components	Months 2–3
3	Temporal Synchronization Module Development	Months 3–4
4	Rule-Based Fusion Engine Implementation	Months 4–5
5	Bayesian Fusion Model Training & Integration	Months 5–6
6	Alert Buffer & Output Publisher Development	Month 6
7	System Integration & Cross-Component Testing	Months 6–7
8	Field Validation & Performance Evaluation	Months 7–8
9	Documentation & Final Submission	Months 8–9

8. RISK ANALYSIS

Risk	Impact	Mitigation
Upstream API schema changes	High	Define stable interface contracts early; version all APIs
Poor sensor time synchronization	High	Use NTP on all ESP32 nodes; implement drift correction
High false positive rate	High	Iterative threshold tuning; Bayesian model retraining
Backend latency exceeding 2s	Medium	Optimize fusion pipeline; use async processing
Insufficient paired training data	Medium	Use simulation and data augmentation for rare events
Sensor node connectivity loss	Medium	Implement graceful degradation with single-modal fallback

9. REFERENCES

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